

In Appendix F we present additional experiments to examine whether "\n\n" causes formation of plans in particular. We find evidence that it can in certain cases, but does not do so generally.

7 DISCUSSION

We have investigated how language models encode and utilize paragraph-scale planning by probing their residual streams at transition points. Our paragraph-level Residual Stream Decoder methods, Continuation ParaScope and TAE ParaScope, reveal that models carry significant information about upcoming content, comparable to having a limited "lookahead" of a 5-10 tokens, and find evidence for the Decodability Hypothesis. We additionally expand the method to a TAE Outline Residual Stream Decoder to probe for longer-context data, and found weak limited results, suggesting there may be some planning, but that it is also likely to be limited in Llama 3.2 3B.

More specifically, we found a linear TAE ParaScope model can often capture broad semantic signals from residual stream activations, such as topic or subject domain, while the Continuation ParaScope is more inconsistent, but offers greater textual coherence in the generated paragraphs and can sometimes reveal more specific details. Together, these findings shed light on the hidden mechanisms of paragraph-scale planning in large language models.

We additionally used Continuation ParaScope to investigate distribution of planning-related signals, and found it is not uniform across all layers. Instead, we see that the middle layers (roughly 60 – 80% of the model’s depth) concentrate these signals, in line with previous research (Meng et al., 2022). Earlier layers focus on local processing and later layers finalize immediate token decisions, whereas the middle layers integrate context to guide broader text generation.

Furthermore, when observing activations at different tokens, we find a sharp shift at paragraph boundaries: the model of study typically creates or refines its "plan" precisely when it processes the \n\n token. This myopic behavior suggests a form of just-in-time updating creation of plans.

These results combine to start forming a picture of how planning behavior in LLMs works.

7.1 FUTURE WORK AND LIMITATIONS

We have so far focused on validation of various methods for Residual Stream Decoding, and found evidence for the Docodability Hypothesis. Since we use two different methods, and run additional tests with Continuation ParaScope, there is some evidence to rule out that the planning information we extract is completely spurious. However, there is room for further research into understanding the degree to which the planning signals we find are strictly causal to model performance.

Additionally, we have focused on short structured paragraphs and full-output outlines. These are not guaranteed to be the most natural scales when it comes to investigating models, and does not generalize cleanly to domains of math, code, and chemistry. It may be the case that there are other units of text, such as sentences, which are more "natural" to probe for. It may also be the case that planning information may be distributed across many tokens, and may not always be cleanly represented in one token.

Finally, our tests are primarily limited to only one model, Llama 3.2 3B, and future work would investigate the degree to which planning differs in models of various different sizes.

Nevertheless, our results emphasize that large language models do maintain recognizable paragraph-level plans — albeit mostly confined to the immediate next section — and that these signals can be partially decoded. Future work may extend this approach to different horizons, investigate how these mechanisms interact with factual correctness or stylistic consistency, and refine interpretability techniques to further clarify the planning processes within large-scale transformer architectures.

These approaches may also lay the groundwork for tools such as monitoring of models in advance of writing an output, and may help in understanding a model’s original intentions prior to completion of an output. We leave these applications to future work.

REFERENCES

- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1): 207–219, 2022.
- Nikhil Bhendawade, Irina Belousova, Qichen Fu, Henry Mason, Mohammad Rastegari, and Mahyar Najibi. Speculative streaming: Fast llm inference without auxiliary models. *arXiv preprint arXiv:2402.11131*, 2024.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/forum?id=ETKGuby0hcs>. Introduces Contrast-Consistent Search (CCS).
- Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, Ryan Greenblatt, Jenny Nitishinskaya, Ansh Radhakrishnan, Fabien Roger, Buck Shlegeris, and Nate Thomas. Causal scrubbing: a method for rigorously testing interpretability hypotheses. AI Alignment Forum, December 2022. Authors sorted alphabetically.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv e-prints*, pp. arXiv–2407, 2024.
- Paul-Ambroise Duquenne, Holger Schwenk, and Benoît Sagot. Sonar: sentence-level multimodal and language-agnostic representations. *arXiv preprint arXiv:2308.11466*, 2023.
- Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 1(1):12, 2021.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. OpenAI Technical Report, 2024.
- Anirudh Gattani, Simran Singh, Mukul Kumar, Tushar Bansal, and Sumit Chandra. RUBICON: Rubric-based evaluation of domain-specific human-AI conversations. In *Proceedings of the 46th International Conference on Software Engineering (ICSE)*, 2024.
- Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. *arXiv preprint arXiv:2401.06102*, 2024.
- Erik Jenner, Shreyas Kapur, Vasil Georgiev, Cameron Allen, Scott Emmons, and Stuart J Russell. Evidence of learned look-ahead in a chess-playing neural network. *Advances in Neural Information Processing Systems*, 37:31410–31437, 2024.
- Gyeonghun Kim, Seonghyeon Park, Jamin Jang, Yoonsik Kim, Donghyeon Shin, Seokmin Kim, and Minjoon Kim. Prometheus: Inducing fine-grained evaluation capability in language models. In *International Conference on Learning Representations (ICLR)*, 2024a.
- Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. Prometheus: Inducing fine-grained evaluation capability in language models. In *ICLR*, 2024b.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kiniment, Nate Rush, Sydney Von Arx, et al. Measuring ai ability to complete long tasks. *arXiv preprint arXiv:2503.14499*, 2025.
- Reagan J. Lee, Samarth Goel, and Kannan Ramchandran. Quantifying positional biases in text embedding models. *arXiv preprint arXiv:2412.15241*, 2024.
- Margaret Li, Weijia Shi, Artidoro Pagnoni, Peter West, and Ari Holtzman. Predicting vs. acting: A trade-off between world modeling & agent modeling. *arXiv preprint arXiv:2407.02446*, 2024.
- Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermy, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language model. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.

- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-eval: Nlg evaluation using gpt-4 with better human alignment. *arXiv preprint arXiv:2303.16634*, 2023.
- Anton Lozhkov, Loubna Ben Allal, Leandro von Werra, and Thomas Wolf. Fineweb-edu: the finest collection of educational content, 2024. URL <https://huggingface.co/datasets/HuggingFaceFW/fineweb-edu>.
- Samuel Marks, Johannes Treutlein, Trenton Bricken, Jack Lindsey, Jonathan Marcus, Siddharth Mishra-Sharma, Daniel Ziegler, Emmanuel Ameisen, Joshua Batson, Tim Belonax, et al. Auditing language models for hidden objectives. *arXiv preprint arXiv:2503.10965*, 2025.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/6f1d43d5a82a37e89b0665b33bf3a182-Paper-Conference.pdf. Introduces Rank-One Model Editing (ROME).
- nostalgebraist. Interpreting gpt: the logit lens. LessWrong, August 2020. Blog post.
- Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022. doi: 10.48550/arXiv.2209.11895.
- Anonymous or Various. Quantitative llm judges. *arXiv preprint arXiv:2506.02945*, 2025.
- Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. *arXiv preprint arXiv:2311.04897*, 2023.
- Alexander Pan, Lijie Chen, and Jacob Steinhardt. Latentqa: Teaching llms to decode activations into natural language. *arXiv preprint arXiv:2412.08686*, 2024.
- Guilherme Penedo, Hynek Kydlíček, Anton Lozhkov, Margaret Mitchell, Colin A Raffel, Leandro Von Werra, Thomas Wolf, et al. The fineweb datasets: Decanting the web for the finest text data at scale. *Advances in Neural Information Processing Systems*, 37:30811–30849, 2024.
- Nicholas Pochinkov, Angelo Benoit, Lovkush Agarwal, Zainab Ali Majid, and Lucile Ter-Minassian. Extracting paragraphs from llm token activations. *arXiv preprint arXiv:2409.06328*, 2024.
- Amy Pu, Hyung Won Chung, Ankur P Parikh, Sebastian Gehrmann, and Thibault Sellam. Learning compact metrics for mt. In *Proceedings of EMNLP*, 2021.
- Mohammad Tafseeque, Philip Quirke, Maximilian Li, Chris Cundy, Aaron David Tucker, Adam Gleave, and Adrià Garriga-Alonso. Planning in a recurrent neural network that plays sokoban. *arXiv preprint arXiv:2407.15421*, 2024.
- Anthropic Interpretability Team. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits report, 2023.
- Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivi re, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- LCM team, Lo c Barrault, Paul-Ambroise Duquenne, Maha Elbayad, Artyom Kozhevnikov, Belen Alastruey, Pierre Andrews, Mariano Coria, Guillaume Couairon, Marta R. Costa-juss , David Dale, Hady Elsahar, Kevin Heffernan, Jo o Maria Janeiro, Tuan Tran, Christophe Ropers, Eduardo S nchez, Robin San Roman, Alexandre Mourachko, Safiyyah Saleem, and Holger Schwenk. Large concept models: Language modeling in a sentence representation space, 2024. URL <https://arxiv.org/abs/2412.08821>.
- Ian Tenney, Patrick Xia, Berlin Chen, Alex Wang, Adam Poliak, R. Thomas McCoy, Najoung Kim, Benjamin Van Durme, Samuel R. Bowman, Dipanjan Das, and Ellie Pavlick. What do you learn from context? probing for sentence structure in contextualized word representations. In *International Conference on Learning Representations*, 2019.

- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv e-prints*, pp. arXiv-2308, 2023.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- Kevin R Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small. *arXiv preprint arXiv:2211.00593*, 2022. doi: 10.48550/arXiv.2211.00593.
- Wilson Wu, John X Morris, and Lionel Levine. Do language models plan ahead for future tokens? *arXiv preprint arXiv:2404.00859*, 2024.
- Xinyi Wu, Yifei Wang, Stefanie Jegelka, and Ali Jadbabaie. On the emergence of position bias in transformers. In *Proceedings of the 42nd International Conference on Machine Learning*. PMLR, 2025.
- Yongjing Yin, Junran Ding, Kai Song, and Yue Zhang. Semformer: Transformer language models with semantic planning. *arXiv preprint arXiv:2409.11143*, 2024.
- Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Joseph E Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023a.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, 2023b.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A DATASET GENERATION DETAILS

A.1 NEXT-PARAGRAPH PREDICTION

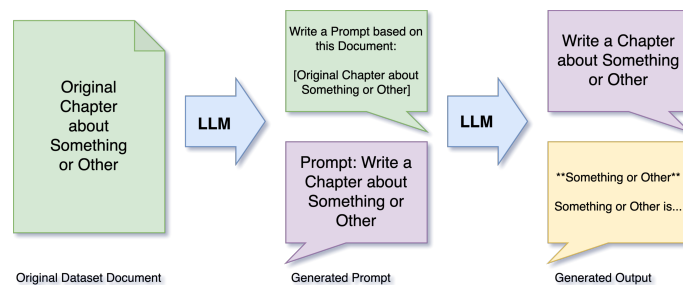


Figure 11: Multi-step dataset generation process.

First, we extract chunks from FineWeb Edu (Penedo et al., 2024), and use Gemma 2 27B (Team et al., 2024) to convert these into structured writing prompts of the form "Write a [type], titled [name], which includes [topics], approximately [length]". This intermediate step helps ensure diversity in our final dataset. We do this for 1 million text samples.

Here is the prompt used to generate the dataset of generated prompts:

```

Write a prompt based on the above text, that is a single-paragraph,
high-level description. Make the prompt in the format format similar to:
"Write a (news feed/chapter/piece/article/wiki entry/...),
  
```